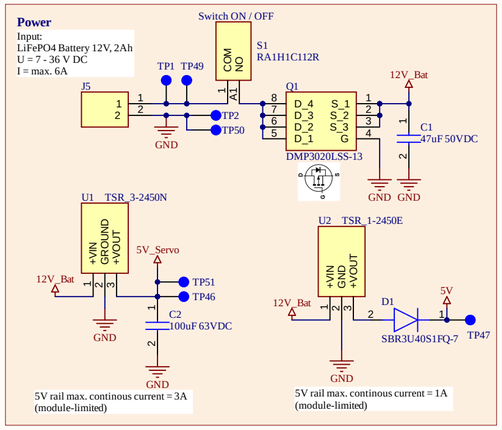
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Power

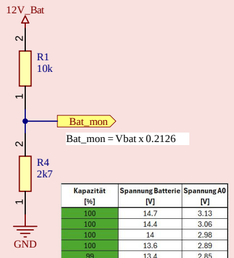
Input:
LiFePO4 Battery 12V, 2Ah
U = 7 - 36 V DC
I = max. 6A



5V rail max. continous current = 3A (module-limited)

5V rail max. continous current = 1A (module-limited)

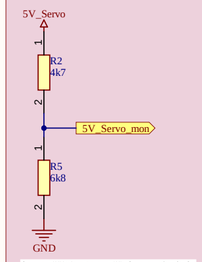
Battery Monitoring



Bat_mon = Vbat x 0.2126

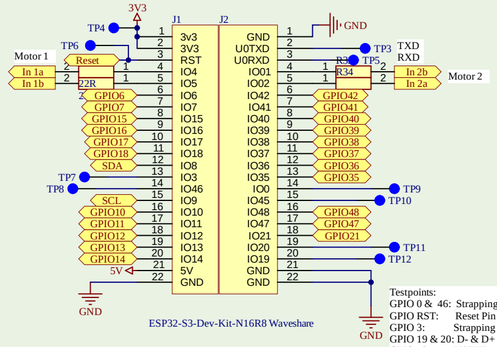
Kapazität [%]	Spannung Batterie [V]	Spannung A0 [V]
100	14.7	3.13
100	14.4	3.06
100	14	2.98
100	13.6	2.89
99	13.4	2.85
90	13.3	2.83
70	13.2	2.81
40	13.1	2.79
30	13	2.76
20	12.9	2.74
17	12.8	2.72
14	12.5	2.66
9	12	2.55
0	10	2.13

5V- Servo monitoring



Servo 5V Monitor = 5V_Servo x 0.5913

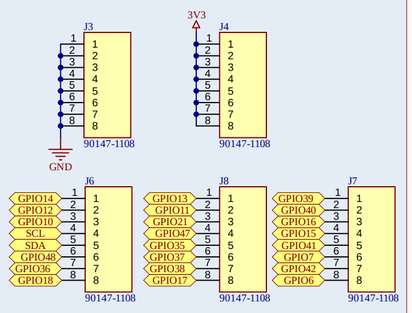
ESP32-S3 Interface



ESP32-S3-Dev-Kit-N16R8 Waveshare

Testpoints:
GPIO 0 & 46: Strapping Pin (Chip boot mode)
GPIO RST: Reset Pin
GPIO 3: Strapping Pin (JTAG signal source)
GPIO 19 & 20: D- & D+ (USB)
GPIO 43: U0TXD serial data from ESP to test device
GPIO 44: U0RXD serial data from test device to ESP
GPIO 45: Strapping Pin (VDD_SPI)

Headers: GPIOs, 3.3V and GND

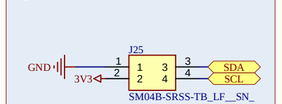


I2C: GPIO 8 (SDA) & 9 (SCL) default I2C as Arduino

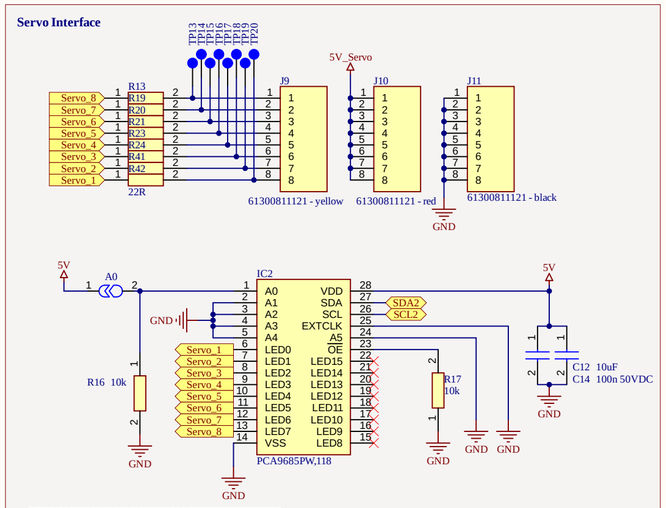
HSPI (SPI 2): GPIO 11 (MOSI), GPIO 13 (MISO), GPIO 12 (CLK), GPIO 10 (CS)

Quad SPI for PSRAM (Webserver): GPIO35/GPIO36/GPIO37

QWIIC - Connector

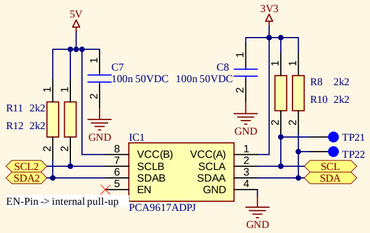


Servo Interface



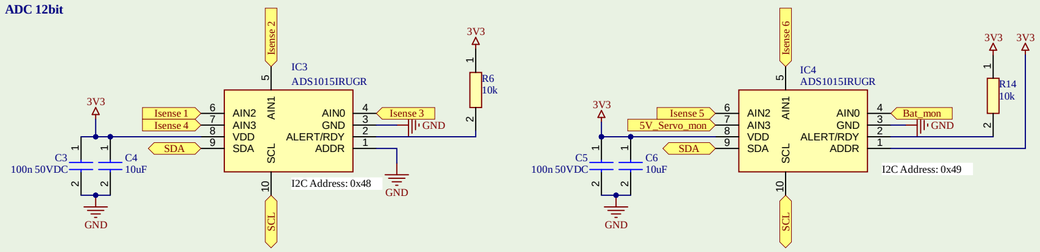
Base Address I2C: 0x40 (binary 0000 no jumpers required)
-> for 0x41 (binary 0001, bridge A0)

I2C Level Shifting

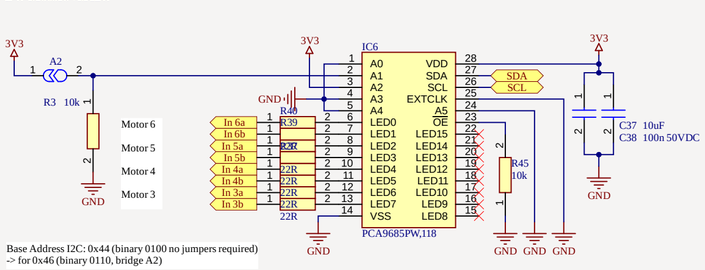


EN-Pin -> internal pull-up

ADC 12bit

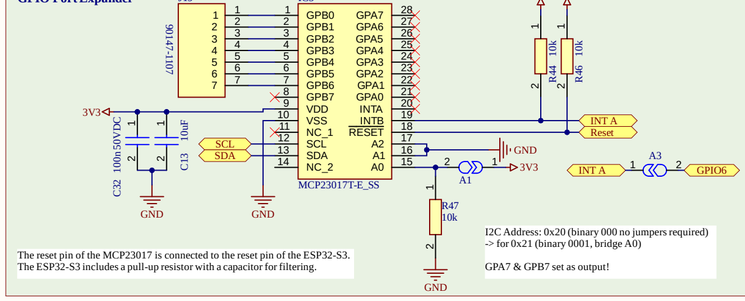


DC-Motors via I2C



Base Address I2C: 0x44 (binary 0100 no jumpers required)
-> for 0x46 (binary 0110, bridge A2)

GPIO Port Expander



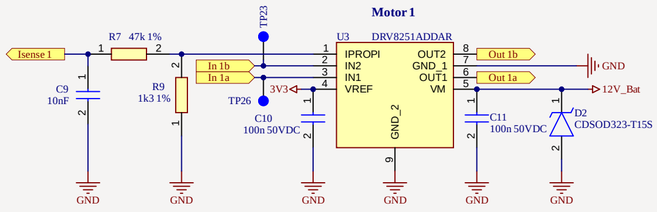
The reset pin of the MCP23017 is connected to the reset pin of the ESP32-S3.
The ESP32-S3 includes a pull-up resistor with a capacitor for filtering.

I2C Address: 0x20 (binary 000 no jumpers required)
-> for 0x21 (binary 0001, bridge A0)

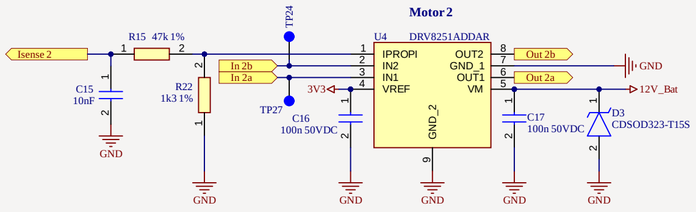
GPA7 & GPB7 set as output!

DC-Motors

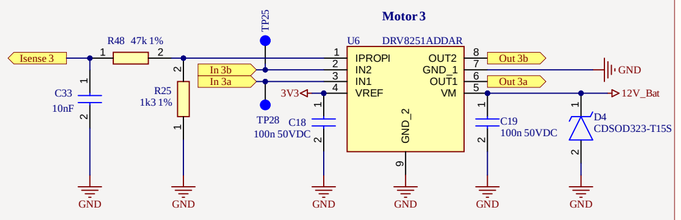
Motor 1



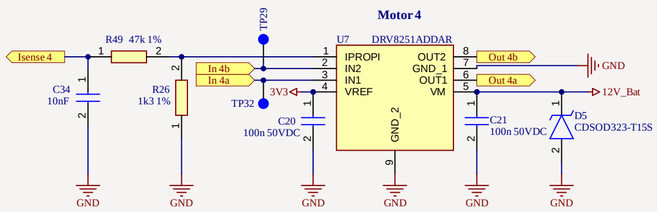
Motor 2



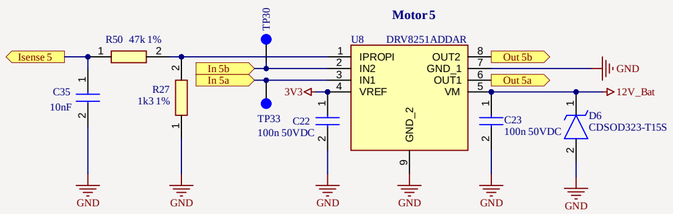
Motor 3



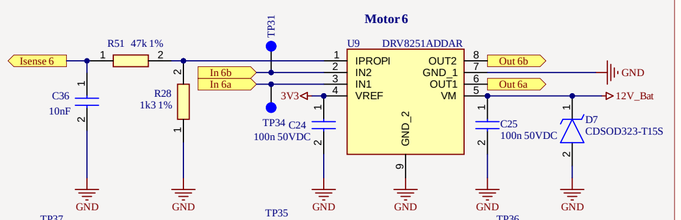
Motor 4



Motor 5



Motor 6

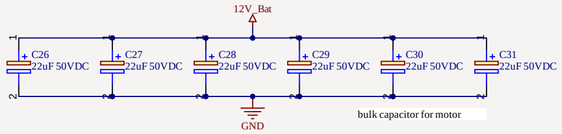


Isense low pass Filter:
tau = R x C = 47k x 10nF = 470us -> 2.1kHz

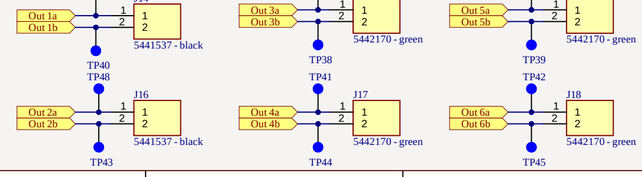
RIpropi = 1.3kohm for current regulation

from datasheet:
max. Ipropi = 3mA
AIpropi = 1575uA
--> Vpropi = Imotor x AIpropi x Rpropi
--> max. Imotor = 1.6A

TVS-Diode (unidirection):
Working Voltage: 15V
Clamping Voltage: 22V
Breakdown Voltage: 16.5V



bulk capacitor for motor



ETH zürich pd|z

DATE 22.07.2026	VERSION V4.0
TITLE	ESP32-S3 DRV8251A Motorshield
SIZE A2	Drawn by: Sara Mettler
	SHEET 1 OF 1